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Conceptualising and exploring creative pedagogies and design thinking in transdisciplinary STEAM higher education courses

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ABSTRACT

Higher Education Institutions are increasingly facing challenges in designing courses that enable students to develop the transdisciplinary knowledge, skills and understanding needed to engage with complex 'wicked' problems. This article reports on a conceptual framework for STEAM in Higher Education (HE), in which design thinking is interwoven with four key features of creative pedagogies: transdisciplinarity, embodied dialogue, empowerment and agency, and ethics and trusteeship. The study explores: how does the design thinking–creative pedagogies framework manifest for participants in practice? The framework was used in four international HE short courses entwining sciences, arts and entrepreneurship to facilitate participants' responses to a different 'wicked problem' in each country, based on the local context. The framework was analysed across the courses to explore how it operates in practice through iterative developments, to help educators understand how to develop it further. The study used a mixed-method, participatory research design, with 82 participants in total. This paper highlights the key creative pedagogy features and design thinking approach working together in productive patterns, helping teachers and partners establish new learning strategies and offering insight into how transdisciplinary learning around complex problems can be facilitated in STEAM Higher Education.

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STEAM; higher education; transdisciplinary; creative pedagogies; wicked problems

Introduction

With global recognition of so-called 'wicked problems' (Kolko 2012), there are increasingly arguments for more transdisciplinary approaches to tackling them (Thomassen and Stentoft 2020). Alongside this, Higher Education Institutions (HEIs) are recognising that they need to build transdisciplinarity and creativity into both research (McWilliam, Hearn, and Haseman 2008) and pedagogy (McGregor 2017) to prepare students to

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respond to wicked problems in their future work and lives (Yeung et al. 2021). STEAM (Science, technology, engineering, arts and mathematics) approaches are one way in which this is being explored in both education and educational research in inter- and trans-disciplinary ways. STEAM approaches often recognise that science education must incorporate learning that can extend to the needs of citizenship, or digital competencies (Ertas 2022; Holbrook, Rannikmäe and Soobard 2020). In advocating for STEAM, it has also been argued that alongside supporting economic growth through engagement of young people with STEM subjects, STEAM can go beyond this, reconfiguring different disciplines to support learning and engagement for educational purposes other than economic drivers (Colucci-Gray et al. 2019). Within this context the arts are seen as enmeshed with, rather than an addition to, STEM (Burnard and Colucci-Gray 2020), with creative pedagogy bridging all disciplines. This approach to STEAM and creativity pedagogy is one we see as 'transdisciplinary', in which subjects are brought together to interact creatively in a learning context as needed with respect to a question or problem at hand. Against this background, the initiative outlined in this paper aimed to conceptualise, implement and study, through participatory research, an innovative approach to HE that could tackle issues faced by educators within STEAM transdisciplinary learning environments which draw on creative pedagogies to prepare students to engage with complex problems such as the future of education, climate change education and the wellbeing of cities.

Literature review and conceptual framework

The term 'wicked problem' is used to describe issues, such as climate change and educational inequalities, that at first appear too complex to address, but that through deep transdisciplinary analysis and navigation can start to be tackled (Thomassen and Stentoft 2020). Recently, there has been a demand for transdisciplinary approaches in HE to address wicked problems (Khoo et al. 2019). Furthermore, STEAM education has been seen as a potential means of dealing with wicked problems in the 21st Century by approaching STEM subjects with an ethical mindset informed by the Arts (Taylor 2016) especially in relation to the issues stemming from the Anthropocene (de la Garza and Travis 2019). STEAM education has also been observed helping to resolve tensions in transdisciplinary learning approaches (Mejias, Gutiérrez, and Vakil 2021) meaning that wicked problems can be approached in a way that avoids conflict in order to ensure learning moves forward in a productive and effective way.

Transdisciplinarity is variously defined in STEAM within HE practice (Baalen et al. 2021): For example, in STEAM education studies it has been linked to Nicolescu's (1997, 4) description of a transdisciplinary space as 'at once *between* the disciplines, *across* the different disciplines, and *beyond* all disciplines' (Liao 2016, 44); '"a new form of learning and problem solving involving cooperation among different parts of society and academia in order to meet complex challenges of society" (Häberli 2000, Shin et al. 2006, 1, cited in Baalen, de Groot, and Noordegraaf-Eelens 2021); and in relation to Ramadier's (2004, 424) definition which states that: 'Transdisciplinarity essentially concerns the articulation between disciplines, rather than their relations, as is the case with pluri- and interdisciplinarity' (Muller and Flohr 2016, 103) amongst others. A concise way to work with these wider definitions, that it is 'an integrated approach to complex

problems using the methodology and insights from a range of disciplines with differing perspectives on the problem under consideration' (Benatar 2000, 171). Within Benatar's work, there is a continuum between, multi-, inter- and trans-disciplinary. Transdisciplinarity is the most enmeshed along the continuum as interaction transcends individual disciplines; it is this core element of the definition that resonates strongly with our understanding of the term within STEAM education and we thus define transdisciplinarity through the relationship between disciplines serving the question or problem in hand, rather than disciplines serving each other (Benatar 2000).

When exploring the description of creativity within STEAM, inconsistencies arise in how it is defined and viewed in relation to research and pedagogy (Colucci-Gray et al. 2017). Often, creativity is defined in relation to singular or plural arts practices, which causes confusion as to how to apply it in different settings (Perignat and Katz-Buonincontro 2018). For STEAM education to be able to be understood and evolve in practice, a model is required that enriches creativity through critical thinking and problem-solving (ibid). Therefore, we have ourselves spearheaded research into creativity in STEAM education, in which creativity is defined as driven by embodied dialogue (Chappell et al. 2019). We argue that new knowledge is created through dialogic interactions between voices of all kinds, and therefore understand creativity as the generation of novel outcomes that are valuable to the learners fostered through dialogue in which learners are engaging with 'others' (including humans and other-than-humans) to view the world from a different perspective (Chappell et al. 2019). These dialogic approaches are positioned within a cohesive conceptual framework which takes a democratic view of creativity in education as an everyday occurrence, rather than a possession of a unique ability (Craft, Jeffrey, and Leibling 2001). We understand that teaching for creativity and teaching creatively are distinct processes which, when actioned together, create a learning environment which is led by the student (Jeffrey and Craft 2004).

Thus positioned, we have worked to both develop and research new STEAM HE pedagogic practice. Our understanding of 'pedagogy' is grounded in a socio-constructivist (Vygotsky 1978) and material (Barad 2007) relationship between the lecturer, learner and environment and therefore encompasses discussion of both teaching and learning. For the purposes of this project, we have also found understandings of creative process and related pedagogy from design thinking very useful. Van der Sanden and Wehrmann (2021) have argued that the double diamond process can facilitate participants to respond to wicked problems by taking them through a process of *discovering* (divergent thinking – a range of possible ideas); *defining* (convergent thinking – refining and narrowing down) in order to find the essence of their response to the problem in hand; *developing* (divergent thinking) and *delivering* (convergent thinking) a shared experience or presentation (for peers and partners) by the end of their process. Van der Sanden and Wehrmann (2021) go on to argue that the design thinking process has the capacity to facilitate a common understanding and shared research process. They argue that participants can be encouraged through 'looped thinking' (Van der Sanden and Wehrmann 2021), which involves the continuous braid of educational theory and practice, science and society and logic and intuition. This cyclical process, combining various strands of thinking, is characterised as complex and ambiguous, and has creativity and critiquing at its heart (Van der Sanden and De Vries 2016). This makes it ideal to be used for STEAM education. Although design thinking in STEAM education has

been recognised as a framework for problem-solving across science and arts (Jantakun, Jantakun, and Jantakoon 2021); as creating a natural bond between arts and science (Henriksen 2017) and as being seen as walking hand-in-hand for change making (Doran et al. 2021), it has not, to our knowledge, been used and examined in STEAM education in HE contexts. By proposing design thinking as at the heart of creative pedagogy in the project described in this paper, we offer a contribution to the field as a newly devised conceptual framework for STEAM in HE. Rooted in Chappell's (2018) embodied, collaborative and dialogic theorisation of creative pedagogy, our conceptual framework is understood to be enmeshed with embodied dialogue through a fluid, ongoing relation between differing entities. As such, the exploration of this framework in practice opens up new ways of thinking about STEAM education because we are interested in the diverse embodied dialogic interactions that take place during the double diamond process rather than the outcomes of each part of it. Also, as we see this as a fluid process, we acknowledge that boundaries are blurred between each part of this process due to the overlapping of embodied dialogue during the sessions. This means that attempting to explain each outcome of the process is too complex for this paper.

With design thinking containing the creative process inherently within it, it was important to consider how best to frame the facilitation of this creativity through creative pedagogies. Theoretically (Chappell et al. 2019) and practically, we have considerable experience in researching and enacting creative pedagogies in varied educational settings to nurture creativity and have previously proposed a set of eight creative pedagogies: dialogue; transdisciplinarity; individual, collaborative and communal activities for change; balance and navigation; empowerment and agency; risk, immersion and play; possibilities; ethics and trusteeship (Chappell et al. 2016). These began as 12 creative pedagogies developed to frame the CREAT-IT European-funded project (implementing creative strategies into formal science teaching). They were synthesised from a creative pedagogies literature review, a survey of 130 European educators, and pedagogic case studies of three creative education projects (Craft, Gardner, and Claxton 2008). Across this project, the twelve features' efficacy and impact were researched using surveys and interviews and they were honed to eight features (Chappell et al. 2015). The CREATIONS European-funded project (developing engaging science classrooms through creative arts approaches) refreshed and updated the pedagogies through a further literature review and workshopping across Europe with 220 educators resulting in the final eight features above (Chappell et al. 2016).

For implementation within the HE STEAM transdisciplinary courses in question it was decided that four creative pedagogies were key, due to them emerging as most prominent within previous STEAM projects where the eight features of creative pedagogy were the main focus of the research (Chappell et al. 2015):

- Transdisciplinarity is defined by the relationship between disciplines serving the question or problem in hand, rather than disciplines serving each other (Benatar 2000).
- Dialogue is pedagogically about asking questions in a way which leads to new ideas and then more questions, with dialogue possible between all kinds of embodied others (Bakhtin 1986; Merleau-Ponty 1964). We understand these emergences as taking place in what Wegerif (2010) refers to as a dialogic space where perspectives

can be created and shifted and new ways of looking at the world are constructed (K. A. Chappell et al. 2012).

- Empowerment and agency is about giving participants ownership of the learning; creating space for them to ask their own questions, make their own mistakes and act with intention. From Barad (2007), we also understand environments as agentic, entangled and emergent. We argue that these agentic environments lead to empowerment.
- Ethics and Trusteeship involves considering the implications and impacts on those around them of their creativity. Trusteeship involves thinking about who holds the values; and within education helping participants to understand that they are the trustees of their own community values, now and in the future (Craft, Claxton and Gardner, 2008).

Thus, within this paper, we are endeavouring to understand how the framework of these four key creative pedagogies, in combination with design thinking, function when applied and developed within SciCulture STEAM HE short courses, and to understand the implications of this for STEAM HE pedagogy. Our research question is therefore: How does the SciCulture conceptual framework manifest in practice in the teaching of transdisciplinary STEAM HE?

Course design

At the heart of this study were four creative, transdisciplinary STEAM HE intensive week-long courses which drew on the conceptual framework described above to connect the arts, sciences and entrepreneurship with design thinking to facilitate participants to engage with and respond to a selected wicked problem, called the 'course challenge'. This future-focused challenge-based approach, drawing on design education research (Moreira 2019) and inspired by the pedagogical model of the Glasgow School of Art (Mcara and Ross 2021), allows transdisciplinarity to be expressed, with participants collaborating and becoming active agents in their own learning and critically questioning their assumptions with the aim of sharing innovative visions for the future by the end of the course. The courses brought together project partners (lecturers in arts and sciences education, entrepreneurship, design thinking, fine art and science communication) and participants from four European HEIs (in Malta, UK, Norway and Netherlands) and one educational agency (in Greece), to work in trans-disciplinary and cross-country groups, with travel and subsistence for face-to-face courses funded through the project. Participants took part in whole group facilitated sessions and were also divided into 'home' groups, which mixed participants from different countries and disciplines to collaborate to address the wicked problem/challenge. Courses were redesigned for each time, however, the core teaching team remained consistent. We saw these courses as 'experimental' because it was the first time that the project partners had engaged in transdisciplinary practice together in this context, using this specific conceptual framework which we studied and developed through the process. The aim was that the research focus of the project would allow for the evaluation of each course to inform the design of the next course in an iterative process.

Table 1. The four SciCulture courses.

Course	'Challenge' topic	Location	No. of participants
1	The Future of Education	Athens, face to face only	19
2	Climate Change Education	Bergen, face to face only	18
4	Wellbeing of Cities	Rotterdam, online (Dutch participants were able to meet face to face, but the course was delivered fully online)	15
3 & 5	Wellbeing of Cities	Malta, hybrid (primarily face to face, with some participants and tutors attending online due to COVID-19 restrictions)	20

The project initially planned five courses, but due to Covid, the delayed third course was combined with the fifth to create a single, larger course. In this paper, the courses are referred to as C1, C2, C4 and C3&5 (see Table 1 for a brief overview of the themes, locations, mode of delivery and participant numbers for each course). Each course combined the classic double-diamond design thinking process (Drew 2019) with creative pedagogies (Chappell et al. 2016) using the conceptual framework described above. This led to careful structuring and sequencing of the courses to include key short 'expert inputs' positioned deliberately within the divergent and convergent points of the double diamond, alongside facilitated group workshop sessions which drew on the different subject areas at times separately to develop participants' knowledge and at times together to demonstrate and foreground disciplinary inter-relationships, and periods where participants worked independently on the problem in a transdisciplinary way. Group sessions involved participants working in 'home groups' deliberately structured by course facilitators to ensure a mix of nationalities and disciplinary backgrounds in each group. Facilitated workshops drew on the creative pedagogies as appropriate for the position within the double diamond to support participants' activity. Figure 1 shows the overview and sequencing of course 2, which exemplifies the structure of all courses. The courses begin with an introduction to the transdisciplinary 'course challenge' and introduction to design thinking. From the challenge, subject-led input and workshops that are collaboratively designed and led by pairs of partners with different subject expertise lead the students through a double diamond of divergence and convergence of ideas with respect to the challenge: the course is planned such that workshops stimulate divergence or convergence at appropriate junctures. In each taught sessions, partners are deliberate in their use of the features of creative pedagogies (see above) to plan and teach the sessions. For example, the movement workshop uses embodied dialogue through movement and dance to prompt participants' thinking in relation to ideas from science or entrepreneurship linked to the challenge, and the learning science through theatre workshop aims to develop empowerment and agency through performance linked to the challenge, scaffolded using the 5Es of scientific inquiry. Thus, Figure 1 offers an indication of what the courses looked like, bearing in mind the description of how the different sessions are designed and scheduled depending on the conceptual framework. All courses used a similar structure and approach. More details about the specific design of each course, and the overarching approach can be found at www.sciculture.eu. (Wren, Chappell, and Hetherington 2021)

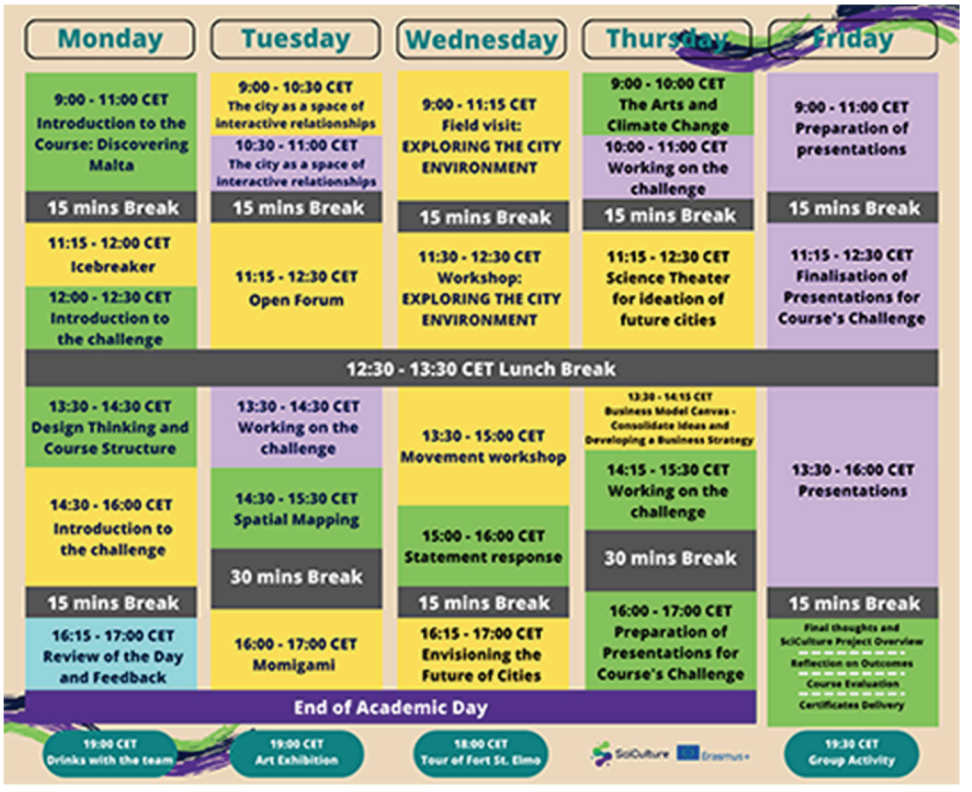


Figure 1. The design and structure of course 2 as an example of the input (green)/workshop (yellow)/home group (purple) structure. The first diamond of the 'double diamond' converges on Wednesday in the 'statement response' session before moving into a further divergent-convergent diamond. Disciplines are connected in workshops, for example in 'exploring the city environment', both science and art techniques are used to collect data and explore, with subject expert lecturers co-designing the sessions to articulate to participants how subjects are in dialogue.

Alongside the use of the conceptual framework in the design of the course, the principles of creative pedagogies and design thinking were used and explained to the participants throughout the course for them to use themselves as a means of addressing a transdisciplinary 'course challenge'. Thus, the conceptual framework is woven through the process of teaching and learning and used by both tutors and participants.

Methodology

A mixed methods approach (Johnson and Onwuegbuzie 2004) was used to gain insight into the manifestation of design thinking and creative pedagogies in the courses under investigation, alongside broader questions regarding the impact of the course on participants' knowledge and understanding that are not the focus of this paper. A mixed methods approach was chosen because it can encapsulate the complexity of the research that quantitative or qualitative research could not capture alone (Creswell 2009); quantitative data offers comparable numeric responses and qualitative data offers insight into

participants' multiple diverse perspectives. In transdisciplinary research projects this methodology has allowed for wide-ranging insights to emerge (Payne 2018; Stinder, Schelte, and Severengiz 2022). Furthermore, this approach can also capture the smaller, less powerful voices and is thus seen, to some extent, as emancipatory (Szostak 2016).

Methods, participants and sampling

A Likert-type pre-post questionnaire was used to gain broad insights into participants' and partners' perspectives via a series of 5-point Likert scales focused on a) participants' confidence in disciplines and transdisciplinarity and b) their knowledge and understanding of design thinking and creative pedagogies (Joshi et al. 2014). These were combined with semi-structured multi-modal focus group interviews (Parker and Tritter 2006) with course participants conducted at the start and end of every course; individual interviews (Powney and Watts 2018) at the end of every course, and a post-project questionnaire with project partners/lecturers; photologs (Richard and Lahman 2015); a reflective mural/padlet to facilitate hybrid digital engagement (Shuker and Burton 2021); and photographs and field observations from teaching and learning sessions and home group end-of-week sharings. Overall, these combined to give a rich insight into each course with opportunity to explore the affective, embodied, everyday experiences (Coffey 2022). Table 2 shows the range of methods used and the number of participants responding to each method. The participant questionnaire was piloted with postgraduate students in England ($n = 48$) before use in C1. Following C1 this was further adjusted to address issues regarding length and complexity of questions. All courses, and data collection, were conducted in English, in which all participants were fluent, with interviews audio-recorded and transcribed for coding.

The research took a stance of ongoing, reflexive ethical practice. Approval for the study was gained from the University ethics board, approval number ST1819–009 taking account of ethics guidance in the UK and EU (BERA 2018; ALLEA – All European Academies 2017), with updated approval sought for methods adjustments in light of the C-19 pandemic move to online/hybrid teaching and data collection.

Participants were postgraduate students and HEI staff from all of the participating countries. They applied through the Erasmus plus grant scheme and were selected for participation by a moderated scoring process in which all project partners took part. Scores were assigned to participants' applications according to their level of interest in the course; how they would benefit from it; experience in Science/Art Entrepreneurship; and transferable skills.

Research was conducted by partner staff and research assistants from three of the participating countries: Malta, UK and Greece, and was coordinated by the UK team.

Analytical approach and procedures

An analytical strategy for all qualitative data was developed and led by the UK team. The first step in analysis was to divide the data set for each course between team members from the UK, Malta and Greece, who conducted an initial deductive thematic analysis in which data was read and examined thoroughly and excerpts coded under key themes derived from the 'features of creative pedagogies' (Chappell 2018) from which the

Table 2. Methods and participants.

Method	Sample	Analysis	Abbreviation of data source.
Participant online pre-questionnaire	72 participants	Pre-Q completed 1 week in advance of each course. Post-Qs completed within 3 days of the end of each course. Following each course, descriptive statistics for ordinal data used to analyse findings. Following the final three courses, results were collated in SPSS and compared using the split file function to look at differences between courses using non-parametric tests and to provide descriptive statistics of an overview of the courses, acknowledging limitations arising due to differences between them.	PreQ
Participant online post-questionnaire	72 participants		Post-Q
Partner online post-questionnaire	10 partners	Completed within 3 days of each course. Analysed using descriptive statistics suitable for ordinal data.	PPost-Q
Participant focus group multi-modal semi-structured interviews on day 1 and day 5 of each intensive course.	4 volunteer participants per course, purposively selected to include representatives of different nationalities, disciplines and experience. 16 participants in total.	Data collected on day 1 and 5 of each course. Analysed using thematic analysis within 4 months of each course (see description in text), with data retained for a final analysis across all courses within 8 months of completion of course 4. Following all 4 courses, codes and themes were re-analysed to explore similarities and consistencies between courses and any overarching themes.	PreFG; PostFG
Partner/staff individual multi-modal semi-structured interviews on day 5 or immediately following each intensive course	10 partners interviewed in total: 6 who took part in all courses, 4 who participated in only 1 or 2 courses.		SI
Fieldnotes using semi-structured schedules, twice daily through each course in lectures or home end-of-week group sharings	All participants and partners (*n*=82)	Field notes were analysed thematically using the same timescale and structure as interview data.	FN
Photography	All participants and partners (*n*=82)	Visual imagers including photographs and reflective activities in mural/padlet were analysed using the same thematic approach as the interview data.	VI
Participant and partner individual reflective photolog; 5 photographs daily; adopted as an adjusted method for C4 online and continued into C3&C5 hybrid.	All participants and partners in C4 and C3&5 (*n*=35)		PL
Participant Reflective Mural/ Padlet activities in C4 and 3&5, adopted as an adjusted method for C4 online and continued into C3&C5 hybrid.	All participants in C4 and 3&5 (*n*=35)		RA

Table 3. Sub-codes related to transdisciplinarity.

Sub-codes	Illustrative data excerpt
building on previous experience	*"In the feedback, one of the groups . . . outlined their take on/interests in transdisciplinarity and education rather than simply things they have in common, [and] in three of the groups' feedback an interest in the intersections between science and art [appeared].* (FN, C2).
promoting constant learning	*"So then if we think this way then I think yes, our scientific knowledge – at least for me – has improved". (PostFG, C1).*
networking	*"The most thing I learnt from working with the diverse [disciplinary group] is think and try to explore the competency and try to [find the] best way to work together." (PostFG, C2, p.2)*
colleagues from different disciplines working towards a shared goal	*"So it's bringing different people together in order to push forward the problem, to get a solution out of it" (PreFG1, C4)*
promoting creativity	*3 words to describe the transdisciplinary experience: 'immersive, challenging and creative'* (SI, C2)

conceptual framework was derived. Each coded data excerpt was labelled by source and type using the abbreviations shown in Table 2. A second stage of analysis was undertaken by the Exeter team to collate the data under the deductive codes and newly emerging themes. This process was undertaken after each of the course, with findings discussed across the project teams and used to inform the next course if needed. Following all four courses, the UK research team undertook a further phase of analysis, collating data from all four courses under the theoretically driven themes, and emergent themes, with careful consideration of whether emergent themes were only arising within specific courses or found across more than one, or all four courses. In this paper, themes reported are those related to all courses. Quantitative data were analysed in SPSS28. Questionnaire data were analysed using descriptive statistics and significance testing appropriate for ordinal data: central tendencies and spread are reported using median and range. Due to the non-parametric nature of the data, non-parametric tests were applied as appropriate to the question being asked of the data (for example, Spearman's rho was used for correlation between variables, and chi-square goodness-of-fit tests to compare frequency of responses between the four courses where the null hypothesis would be no difference in response). No attempt was made to conduct a multiple-regression analysis to identify interactions between elements in the conceptual framework due to the reflexive and evolving nature of the courses. Instead, the analytical approach explored the data about individual courses and between courses in order to layer both quantitative and qualitative data into a description of the courses and the enactment of the conceptual framework. During the analysis of both quantitative and qualitative data, the research team engaged in memo-ing to explain and reflect on their thought processes where relevant (Braun and Clarke 2022).

This process was used to create interim reports on the findings from each course which were used iteratively to improve the next set of courses, drawing on the principles of an action research cycle (Somekh 2005). On completion of analysis of all four courses, the outcomes of this analytical process were synthesised to explore patterns over time in how the courses manifested design thinking and creative pedagogies. It is this over-arching final analysis that is reported in this paper, responding to the research question 'How does the SciCulture conceptual framework manifest in practice in the teaching of transdisciplinary STEAM HE?'.

Analysis and findings

To understand participants' perceptions and experiences of the conceptual framework in practice (which, as described above, was explicitly implemented in the course and explained to the participants as a means of addressing the course challenge as well as being used by the team to design the courses), the analysis presented below begins by unpicking results with respect to the five distinct elements of the framework, namely design thinking, and the four features of creative pedagogies foregrounded in the SciCulture conceptual framework for STEAM HE: trandisciplinarity, creativity, empowerment and agency, and ethics and trusteeship. Where possible, relationships between these aspects of the framework across the four different courses are highlighted, before discussing insights from these findings for refining the SciCulture Conceptual Framework. It should be noted that the analysis is illustrated with a limited set of excerpts from the extensive data-set for reasons of space.

Design thinking

At the beginning of all courses most participants did not really know what design thinking was or how it linked to creativity: 'The first time I heard of design thinking was yesterday' (PreFG, C1.); 'I didn't know anything about design thinking [at the start]' (PostFG, C2) and 'I only had that one experience with design thinking' (PreFG, C4). Participants' understanding of design thinking improved across the courses, with a statistically significant increase in participants' self-reported knowledge and understanding of Design Thinking (pre-median = 3; post-median = 4,$Z = -3.435$, $p < 0.01$), which was also demonstrated in interviews: 'Design thinking I think is basically a practice[. . .] I learned how to design your whole problem or idea and give how, give the whole process in a form of design' (PostFG,C2).

Photolog data Figure 2(a,b) shows participants engaging with and understanding the design thinking double diamond at the heart of the course through embodied dialogue and object sculpture (Figure 2(a), created as participants discussed and explored the idea of design thinking) and self-initiated arrangement of momigami paper[1] with personal annotations (Figure 2(b)). Figure 2(b) is particularly powerful as it was initiated by a participant in their Photolog and shows them clearly focusing in on the design thinking process as part of their reflections.

Participants with less prior experience in transdisciplinarity tended to show a larger change in their confidence in design thinking across the courses (as demonstrated by a negative correlation between prior experience in transdisciplinarity and size of difference in confidence with design thinking before and after the course; Spearman's rho = -0.587, $p = 0.010$). This suggests a relationship between transdisciplinarity and design thinking, as participants who were more experienced in transdisciplinarity had higher confidence in design thinking in pre-course questionnaires.

These findings suggest that design thinking was a core aspect of the course practice that participants used in their learning and reflections across courses as well as being used by partners and course facilitators in the design and implementation of the courses.

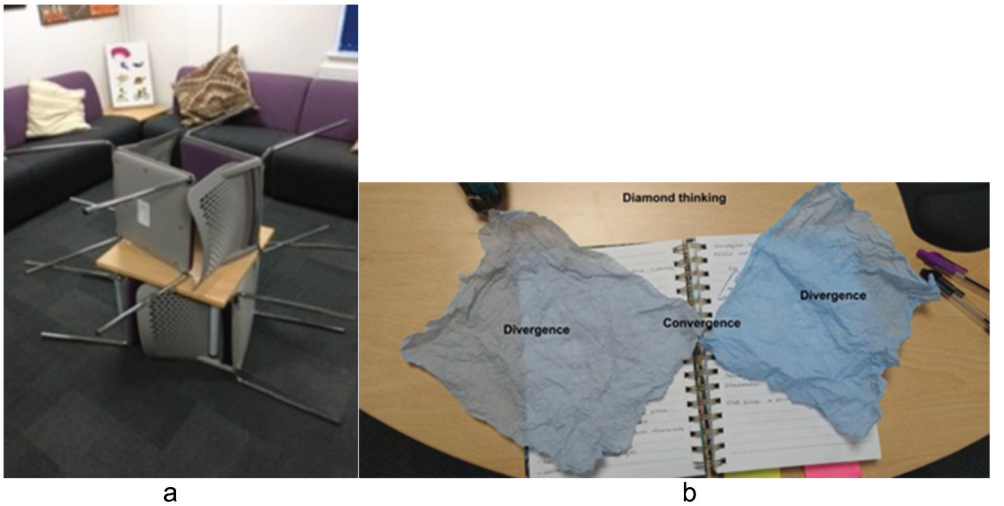


Figure 2. (a,b). Enactment and reflection on design thinking demonstrated by participants in different ways.

Transdisciplinarity

Across all courses, participants were more likely to respond that they made links between disciplines rather than prioritising learning within disciplines (median = 4 compared with median = 3, $n = 62$). However, it should be noted that there was some variability between courses about the extent to which participants felt they were making links between disciplines. C2 showed the highest rating (median = 4.5, $n = 18$). In C4, which was fully online, there was a lower average response (median = 3, $n = 15$), and in C3&C5, the hybrid course, participants' median response to the question 'I made links between disciplines' was 4 ($n = 20$). These differences are corroborated by participants' comments. One explanation of this finding might be that it reflects the shift to fully online and then hybrid formats, with transdisciplinarity thriving most in face-to-face contexts. Additional explanations could include the specific knowledge and understanding of the participants, different course topics or changing approaches from course tutors, but the nature of transdisciplinary learning in face to face or online contexts is certainly worthy of further exploration.

Thematic analysis suggests that participants did not adopt a single definition or perspective on transdisciplinarity following the course. Key themes related to distinctive views of transdisciplinarity included a) focusing on a developmental and iterative perspective, represented by the analytical sub-codes 'building on previous experience' and 'promoting constant learning', b) focusing on collaboration with others, represented by the analytical sub-codes 'networking' and 'colleagues from different disciplines working towards a shared goal', and c) about promoting creativity (see Table 3). One participant's description reflects the way in which transdisciplinarity is used in the SciCulture conceptual framework: *'Different people have studied in different fields, and it gets them together in order to work towards a goal. So it's bringing different people together in order to push forward the problem, to get a solution out of it' (PreFG1, C4)*. This

set of perspectives on transdisciplinarity, linked to participants' reported experience of drawing out more links between disciplines during the courses, demonstrates the broad course aims of fostering learning, collaboration, problem-solving and creativity through transdisciplinarity. The notions of a developmental and iterative approach, and creativity, can also be linked to the double-diamond of design thinking, and shows a qualitative through-line to the connection with design thinking.

The rewards and challenges of transdisciplinarity are evident in the qualitative data: '*I found it challenging . . . at times – different expectations, different backgrounds, different approaches. At the same time I did find it really rewarding when we did, like, come together in something and we perhaps shared or found an insight or built something. And I still think it's really exciting to be working with people from different knowledge fields*' (FG2, C3&C5). Despite the rewards, the data suggests a range of challenges, including finding the balance between the nature of transdisciplinarity (including the very open nature of the problems framing the course) and its manifestation (or otherwise) in the course pedagogies such as the 'direct input' elements of the course design (see Figure 1): '*It is perhaps jarring to go from a very interactive and energetic activity using art and animation, to a lecture on entrepreneurship*' (FN, C1), and the potential for only a surface level exploration during a short course.

In both C4 and C3/5 there is a suggestion that although the courses enabled participants to think about and reflect on transdisciplinarity, it was not always evident in the course practices (for example, in mural reflections on C3/5, participants said they would have liked 'more connection between lectures and the process'. To critically understand this, in C4, a Mural tile indicated use of transdisciplinarity (Figure 3), yet a participant stated: '*I thought it was really interesting seeing all the transdisciplinary perspectives and really thought-provoking seeing them. But . . . I felt they didn't quite mesh, they didn't come together.*' (PostFG, C4). Tutor interviews highlighted the challenges in putting transdisciplinary ideas into practice, particularly when participants are themselves challenged when working outside their normal disciplinary 'comfort zone' '*[the participant said that] he did not understand [my embodied dialogue session] to the point that he didn't know what questions to ask to get access to what [I] was talking about. He described it as sitting with visuals coming into his eyes but he couldn't process it in relation to his own knowledge base' (SI, C1)*. This is a challenge for the implementation of the conceptual framework for STEAM HE in

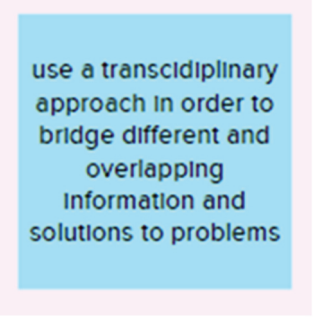


Figure 3. Reflection on challenges of transdisciplinarity, from C4.



Figure 4. Frottage rubbings in the lab.

practice rather than teaching *about* it. Finding the balance between including sufficient information for participants to access the disciplines they are least knowledgeable about, in order to enable them to work in a transdisciplinary way, remains a challenge.

Although the reported participant perceptions did not suggest they saw greater practice of transdisciplinarity in each course, partners felt more confident in the developing transdisciplinarity as they continued to develop the courses over time, albeit somewhat disrupted by the move online for C4 and to hybrid for C3/5. This was facilitated by a decision for partners from different disciplines to co-plan and teach sessions from C2 onwards as a result of the evaluation of C1. *'[The] transdisciplinary element also feels stronger within the student work, I don't know whether that's because of the co-leading by the partners but it definitely feels more connected this time [in comparison with C1]' (SI, C2)*. In C4 partners felt that the transdisciplinarity aspect was more fluid, and in C3&C5 half the partners commented on a particularly successful run of transdisciplinary facilitation when a science/fine arts lab had flowed into a movement installation workshop around pollution and human response to it. Figure 4 shows Frottage rubbings in the lab alongside pollution detectors, illustrating this transdisciplinary approach:

Although course tutors suggest a stronger sense of transdisciplinary working, the participants' perceptions of transdisciplinarity in practice does not suggest that this always translated into practice, and this is certainly worthy of further critical reflection with respect to the overall conceptual framework and how it is drawn on as part of the course design. The breadth of perspective on transdisciplinarity described in this section (collaboration, creativity and ongoing learning as well as the drawing together of disciplines as needed to respond to a key challenge) and how this connects to the pedagogical approaches chosen needs to be considered as part of the framework and its use.

Dialogue

Embodied dialogue was a key concept in the SciCulture framework, including the notion that dialogue is understood as questions which lead to answers which lead to further questions but goes beyond talking over ideas to include the materiality of the body and environment (C. T. Chappell 2019). Learning through continuous questioning was apparent in all courses (34 participants agreed or strongly agreed regarding this, median = 4, $n = 53$). This was least strongly noted by participants in C4, the online course (C2 median = 4, $n = 18$; C4 median = 3, $n = 15$ and C3&C5 median = 4, $n = 20$), indicating that perhaps dialogic chains of questioning were more challenging to initiate and maintain in an online intensive course.

During the courses, our findings show strong evidence of embodied dialogue, albeit with somewhat differing manifestation, with evidence of dialogue being used to understand difference and navigate barriers, and to connect with 'others' (Bakhtin 1981) but also some discomfort with engaging in dialogue, with respect to embodiment but may also relate in part to learning in a second language. Sub-codes related to embodied dialogue demonstrate that dialogue was used and developed in the courses with respect to themes of a) communication, discussion and the exchange of information b) mutual learning, c) problem-solving, d) embodiment, body language and movement, e) understanding difference and navigating barriers, and f) ethics (see Table 4 for a summary of sub-codes and illustrative data).

The sub-codes identified in the data analysis linked to the embodied dialogue concept in the SciCulture framework are suggestive of relationships with other aspects of the framework in the way the framework was enacted in practice. For instance, problem-solving is a key facet of both the design thinking approach, and the notion of transdisciplinarity as a means of finding of solutions and responses to key challenges is also a point of connection. Similarly, communication and the importance of understanding difference and navigating barriers can be connected to the challenges of transdisciplinarity and the importance of dialogue as a means of addressing these challenges.

With respect to the idea of embodied dialogue specifically, participants in all courses identified bodily interaction and interaction with the material world as important to their learning: *'I had no expectations, but it absolutely blew my mind (and body)' (PostQ, C2)*. Figures 5 and 6 exemplify material and embodied dialogue developed through the course. Unsurprisingly, although present in C4 (Figure 6), this

Table 4. Sub-codes related to embodied dialogue.

Sub-codes	Illustrative data excerpt
Communication, discussion, exchange of information	*"The students actively listened to each other and also communicated their ideas. They shared personal experiences".* (FN, C1).
Mutual learning	*"I liked the fact that I could learn from them and they actually learned from me" (SI, C2).*
Problem-solving	*" . . . all ideas being represented and accepted. And it is a challenge how to present all ideas together, how all ideas are going to work . . . "* (FG, C3/5)
Embodiment, body language and movement	*"When participant next to him speaks, he changes is position. Switches crossed leg and leans torso towards the participant speaking"* (FN, C3/5)
Understanding difference	*"In my groups, because we're from about three different countries, so sometimes we think things from different perspectives but I think throughout those days, the activity, we can combine our opinions about one topic and we can make good communication with each other"* (FG, C1)
ethics	*"Some of them have appeared separate and I'm trying to bring them in"* (SI, C2).



Figure 5. Embodied dialogue; participants physically 'tuned in' to each other.



Figure 6. Reflections from C3&5.

aspect was comparatively more strongly rated in C2 and C3/5 (median = 4 for statements related to learning through bodily interaction and material interaction in both C2 and C3/4), with a statistically significant difference between the online course, C4, and the other courses with respect to the questionnaire statement focused on embodiment (median = 2, $n = 15$; Chi-square = 12.556, $p < 0.005$). Interestingly, this difference was not significant with respect to *material* rather than bodily interaction for C4 (median = 3, $n = 15$), which implies that the partners' efforts to include multimodal dialogue and prompt material interactions as part of the dialogue was to some extent successful. Issues were raised by participants in C4 (online) and C3&C5 (hybrid) where the digital platform prevented or interrupted dialogue in places (see Figure 7), though in C3&C5 absences of speech during the final 'performance' of the course outcomes refocused attention to bodies: *'There is an oddly dreamlike quality to being unable to hear all the words in the performance, it forces focus and attention on the bodies and objects intra-acting, lending a sense of bodies in dialogue in the space'. (FN, C3&C5)*.

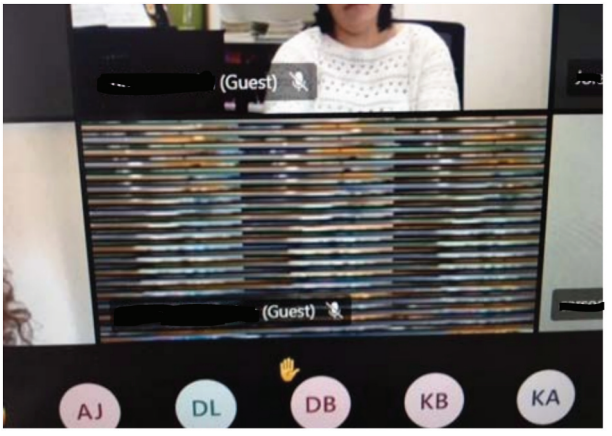


Figure 7. Digital interruption to dialogue (C4, FN).

These findings have interesting implications for the manifestation of the embodied dialogue facet of the conceptual framework in practice during online rather than face to face or hybrid teaching, suggesting that attention be paid to material interactions in using the conceptual framework to design and teach an online STEAM HE course.

Ethics and trusteeship

Overall, participants felt it was important to think about the ethical implications of their work and learning (overall median = 4, $n = 53$, post-questionnaires). In C4, however, the average response to 'it was important to think about ethical implications' was to disagree (median = 2, $n = 15$), though the feature still appeared to emerge within the data, for example in accounting for safety considerations in city planning in C4: the reasons for greater disagreement with this statement are unclear.

In C1, participants were directly encouraged to think about ethics in sessions Figure 8(a,b), but across all courses, participants were also observed thinking and acting ethically, with one describing that the course '*. . .fulfilled an important need to create a network of people who care about the world and want to innovate together*' (PostQ, C1). Participant data suggests some distinctions between 'thinking about ethics' during the course and 'being ethical'. With respect to thinking about ethics,

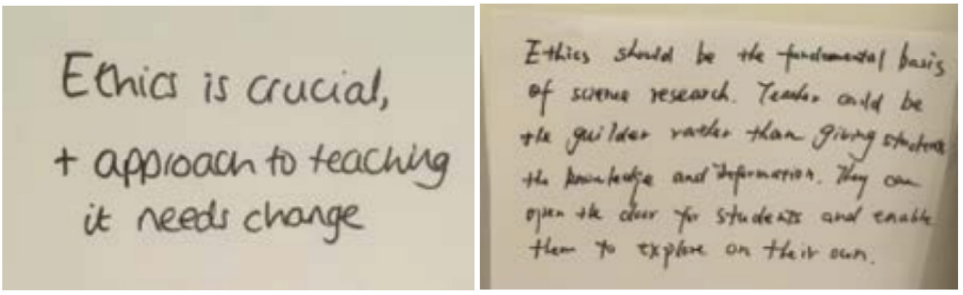


Figure 8. (a,b). Photographs taken of participant post-it notes during a session in C1.

Table 5. Sub-codes related to ethics and trusteeship.

Sub-codes	Illustrative data excerpt
Thinking ethically	*"It fulfilled an important need to create a network of people who care about the world and want to innovate together. I had been longing for such a nourishing environment"* (Post-Q)
Being ethical	
Embodied Trusteeship	
Affect	*[Affected by ideas of future under discussion]: conflicted, hand on face. Seems frustrated"* (FN, C4)
Space and culture	*"I guess it would be nice to focus on maltese culture, maltese space"* (FN, C3/5).
Light	*They reply that it's about natural light. A debate develops about natural and fake lighting that creates the 'wrong' or 'right' atmosphere"* (FN, C3&C5)

there was evidence of discussion about the role of humanity in relation to the world as a philosophical question. 'being ethical' was demonstrated in the participants' responses to each other during points of tension and conflict, as well as in their practical considerations about their responses to course challenges (see Table 5). This suggests that there are varied ethical dimensions at play during the courses.

Emergent codes in the data analysis were connected with the overarching theme of ethics and trusteeship in relation to the specific course challenge. Firstly, with respect to space and culture, there was a consideration that the location of the course had an important part to play in the theme of the wellbeing of cities and consideration of what could and should be an ethical approach to developing a 'greener' city for the future that takes accounts the cultural needs of the city as well as considering climate change: '*I guess it would be nice to focus on Maltese culture, Maltese space*' (FN, C3&C5). A second emergent sub-code with respect to the theme of ethics was 'light and wellbeing', which arose from a debate about the importance of natural light (see Table 5)

With respect to Trusteeship, the role of humanity in relation to others was apparent as a key idea, with at times affective responses seen. During C2, a participant shared '*how pessimistic he feels about the future and the other participant who was quiet the whole time said "I think humans need to be wiped out" with a smile, agreeing that the future is bleak.*' (FN, C2). Some data, such as Figure 10, were coded against both trusteeship and embodied dialogue, indicating a connection between them. In the above section, the importance of dialogue as fostering collaboration and connection with each other was identified and this is also a key part of trusteeship, which could also be viewed as an illustration of the way in which different aspects of the conceptual framework co-exist and intertwine.

Empowerment and agency

In this study, we understand empowerment and agency as a creative pedagogy where participants take ownership of the learning and ask their own questions, make their own mistakes and act with intention (Chappell et al. 2015; Craft, Jeffrey, and Leibling 2001). We saw this feature across all courses and all data types, as exemplified by this excerpt: *'The participants are leading the partners for where to be and what to do in this case' (FN, C3&C5)*. Questionnaire data across all courses shows that participants felt able to both make choices (median = 4, $n = 53$) and enact their choices (median = 4, $n = 53$). Results between the different courses are similar, with no statistically significant differences in participants' perspective between courses. Qualitative data suggests that empowerment and agency was associated by participants and partners (the latter in interviews and field notes) with expanding knowledge, with experiential learning, in voicing opinions (see sub-codes summarised in Table 6), suggesting that it is linked with both dialogue and transdisciplinarity. Interestingly, data suggested empowerment and agency co-manifested in the course with embodied dialogue, such as in C2 and C3&C5 movement and embodied dialogue sessions (see Figures 5 and 9). Participants were seen sharing their ideas to make a representation of their groups' proposed activities. *'They also used the floor while balancing on their hands. At first the rest of participants looked at them with a sense of confusion, but they instantly became more motivated and looser, as they also tried more challenging body postures including having physical contact with each other' (FN, C2)*.

Empowerment and Agency was associated with confusion or dissension (see Table 6), and, unsurprisingly, when a decision to act was required. The connection with periods of dissension, and with points of decision making, link this aspect of the conceptual framework with the 'double diamond' of design thinking: the periods of divergent thinking which could be associated with confusion or dissension, and the decisions needed to shift into a more convergent mode.

As Table 6 shows, the sub-code 'gender' suggested that to some extent the enactment of agency in groups could be gendered, with males dominating discussion. Similarly, disempowerment was evident in the data, as well as empowerment and agency. This is worth further critical interrogation, as where shared responses are ultimately required personal opinions may need to be subjugated; ensuring that this is done appropriately or further space given for individual voice is key to developing practice-based

Table 6. Sub-codes relating to empowerment & agency.

Sub-code	Indicative data excerpt
Rooted in knowledge	*"The new knowledge that they allow us to expand our thinking and get our ideas a step forward." (PostFG, C1)*
Experiential	*".my knowledge was stimulated through every workshop and discussed with the partners and the people in my group, rather than learning according to read new things, search for art stuff, for landscape architecture" (PostFG, C4*
Voicing opinions	*"Different participants throughout the talk have been voicing their opinions and views" (FN, C2)*
Gendered	*"I noticed the men tended to lead the discussions around the table in this session." (FN, C2)*
Confusion or dissension	*"Participant sits to the side of the camera and explains what they need to do. She complains about information overload and that she is overwhelmed" (FN, C4)*
Disempowerment	*"Not everyone communicates in the same way. An analytic person finds it harder to express their opinions". (FN, C1).*



Figure 9. Embodied dialogue and trusteeship during a session focused on climate education (C2).

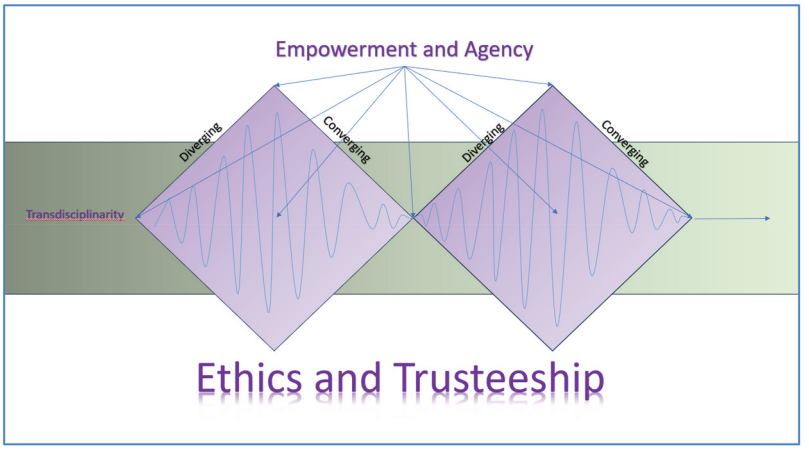


Figure 10. A revised SciCulture conceptual framework.

understanding of this pedagogy. This balance between the individual and the group with respect to agency was not originally articulated within the conceptual framework but is important to consider with respect to the interactions between this element of the framework and others that are clearly focused on collaboration.

Discussion

In working to understand how participants perceived courses designed and taught using the SciCulture conceptual framework for STEAM in HE, these findings offer insight into how the different facets of the conceptual framework interacted in practice. A transdisciplinarity approach was evident in the courses, with participants' describing their developing understanding, and use, of disciplines as needed in responding to the course challenge. This disciplinary fluidity (Thomassen and Stentoft 2020) helped participants respond to the course challenge, or 'wicked problem' they were asked to consider.

However, it is clear that partners, as facilitating lecturers, needed to adapt to some of the challenges which arose, such as participants feeling able to access disciplinary approaches beyond their own prior experience, or enabling participants to recognise and identify the transdisciplinary praxis as well as transdisciplinary learning. To ensure that this transdisciplinary facet of the conceptual framework can be more widely understood and utilised within HE, it may be useful to think of transdisciplinarity as a 'way of being' to extend transdisciplinarity beyond students' learning situations into participants' everyday lives (Rigolot 2020). It may also help to more clearly articulate to HE students each of the features of creative pedagogy within the framework (i.e. embodied dialogue, empowerment and agency etc.) and demonstrate their action in practice, to aid their recognition of and developmental reflection on them. Furthermore, during the design and application of transdisciplinary pedagogy which addresses wicked problems, lecturers need to ensure that the diverse perspectives of everyone involved are heard, and indeed, that the complexity of the issue in hand is well understood (Pohl, Truffer, and Hirsch Hadorn 2017).

Whilst the digital platform emerged as influential on transdisciplinarity and the other features in C4 and C3&5, participants' dialogic questioning of ideas emerged more in C2 than C4 and C3&C5 and there was evidence of technical issues interrupting dialogue. This could be related to how dialogue is or is not able to manifest on a digital platform (Kergel 2018), and the difficulty in switching to the digital quickly due to C-19 lockdowns. In order to address this, Kergel's (2018) suggestion might be applied, of seeing dialogue as pluralistic and diverse interactions which do not seek a universal truth, allowing diverse narratives to emerge in varied media. The relative importance of materiality in comparison with embodiment in C4, the online course, adds a further dimension, layering the importance of material interaction as part of this diversity. Coupled with the evidence of dialogue being used to understand difference in C1, 2 and 4 it is imperative that lecturers in HE are aware of the many different paths dialogue can take both face to face and online, and allow it to take shape despite any glitches that may occur. With understanding difference also seen as a key feature of transdisciplinarity (Klenk 2018), it is important for creative pedagogical design in this context to have dialogue at its heart. Paying attention to the diversity of dialogic interactions, including embodied and material dialogue, might be a way of addressing the highlighted balance between individual and group empowerment and agency. Embodied dialogue was also strongly linked to empowerment and agency which manifested as sub themes relating to requiring an understanding of differences of gendered voices and a variety of knowledge, particularly within the participant cohort. This again illustrates strength in the design of transdisciplinary, embodied approaches to pedagogy by allowing these diverse voices to come through. However, this can be difficult to manage due its complexities. To this end, it is important that lecturers are also empowered to deal with differing cultured and gendered power structures within cohorts, so that courses are run reflectively, professionally and effectively (Whiteside, Tsey, and Cadet-James 2011). Furthermore, this reinforces Sengupta, Shanahan, and Kim's (2019) work which argues that the promotion of critical, transdisciplinary, embodied pedagogy in HE can empower both learners and lecturers to make visible hegemonic structures

which can be collectively reimagined for social justice. With strong evidence throughout the courses of dialogue being experienced through embodiment, these findings also reinforce that, when designing pedagogy for responding to wicked problems in HE, it is helpful for lecturers to understand dialogue within design thinking as multimodal and linked to the lived-in body (Poulsen and Thøgersen 2011).

Whilst ethics and trusteeship was generally identified as important in the use of the conceptual framework in practice, C4 participants disagreed about the importance of ethics, raising questions as to the differences in ethical debate and how it can be fully engaged in on digital platforms. Certainly, related to ethical depth, capturing human emotions in HE digital learning has been highlighted as an issue (Castañeda and Selwyn 2018). It is also likely that C4 participants were suffering from 'zoom fatigue' due to the intensive and short-term nature of the course, which may explain their lack of awareness towards their ethical thinking. Nevertheless, in C3&C5 participants were seen as considering ethics if it was prioritised in the teaching, showing that this can occur within digital environments. With varied creative pedagogy dynamics evidenced within the courses' digital spaces, our findings support Castañeda and Selwyn's (2018) arguments that HE digital research needs to pay greater attention to pedagogy. It is worthy of note though that although trusteeship was evidenced in the courses, it was less apparent than other aspects of the conceptual framework. The reasons for this are unclear from the data. Perhaps trusteeship as a group is challenging to develop over a short-time period such as the week-long intensive courses described in this study. Perhaps it is a difficult concept to grasp, particularly for Western participants for whom it is not traditionally a central philosophical tenet. Further work is needed to understand students' perspectives on the role of trusteeship in transdisciplinary STEAM HE, which we believe is important in terms of students taking ownership and responsibility for responding to wicked problems in their future professional lives (Hoinle, Roose, and Shekhar 2021), related to their empowerment and agency. Our findings highlight the importance of this as a future area of focus to respond to the issues faced by lecturers facilitating transdisciplinary learning around complex problems. This is already a focus of some first nations, non-European research and education, such as the Maori concept of kaitiakitanga, or environmental trusteeship (Ritchie 2013).

Khoo et al. (2019) argue that working in the transdisciplinary arena, pedagogical approaches become more open-ended, pluralistic and complex. Our findings suggest that a complex range of pedagogies were enacted in the courses, rooted in the conceptual framework, but that this also posed challenges in practice particularly in an online environment. Transdisciplinary pedagogies can also become more inclusive through the diverse conversations between different backgrounds, but can be unpredictable especially through digital breakdowns. There are also potential tensions which were not explored in depth in these courses with respect to assessment, particular those that may arise if participants, used to finding answers in more traditional educational approaches, may be challenged by transdisciplinary dialogues where a more fluid outcome might also serve as a quality response. Despite these challenges, the broadly positive responses of participants to the courses described in this study (see Wren, Chappell, and Hetherington 2021) suggests that the conceptual framework can offer an important contribution to those wishing to develop, and research, transdisciplinary HE practice.

Conclusion

Our findings and discussion suggest that the conceptual framework enabled the transdisciplinary STEAM HE courses to be designed effectively. With respect to the conceptual framework itself, the relationships in the data between the different facets of the framework show that, although there are key points in the framework where connections can and should be made explicitly, such as in the role of dialogue in divergent and convergent elements of the design thinking 'double diamond', the relations between design thinking and the core key creative pedagogies within the framework need to be fluid and intertwined. Managing this fluidity is a core skill that facilitators of transdisciplinary STEAM HE need to manage, and it was noted in our data that, possibly especially within a short course, meshing sufficient disciplinary input to support understanding of the core challenge at hand with sufficient openness and fluidity in how disciplines are drawn together was difficult. This fluidity could be achieved through dialogue and empowerment and agency as key creative pedagogies, supported by ethics and trusteeship as underpinning values. Figure 10 shows a refined diagram of the SciCulture conceptual framework for transdisciplinary STEAM HE that, based on these findings, shows the links between key features, taking account of the changing role of embodied dialogue within the double diamond process, the points at which empowerment and agency in the framework are particularly key (at turning points within the double diamond), and ethics and trusteeship situated as underpinning the framework. This development of the framework is significant because, rooted in substantial findings from the SciCulture courses, it offers more detailed guidance for how these theoretically driven features relate to each other, albeit in a fluid and dynamic way, thus offering more substantial and helpful guidance for practice that is based in both theory and empirical study.

It may be the case that if HE pedagogies become more transdisciplinary, different kinds of transdisciplinarity will become apparent. It is therefore important to acknowledge that different transdisciplinary models will work for different lecturers and learners. The conceptual framework proposed here was developed specifically with intensive STEAM HE courses in mind and can be adapted for different transdisciplinary settings. However, the core principles combining design thinking with key features of creative pedagogies (transdisciplinarity, embodied dialogue, empowerment and agency and ethics and trusteeship) may well translate into different blends of course length and disciplines. It has been argued that strong disciplinary backgrounds are necessary so that a transdisciplinary approach can work well (Smith and Kanuka 2018). Our findings with respect to embodied and material dialogue, and space and culture, highlighted the educational benefits of being able to move around in large areas, outside and within different cultures to enhance collaborative learning, empowerment, trusteeship and ethics, also suggest that there is a need for learning spaces, both digital and physical, that can be adapted to account for different embodied and material dialogic interactions.

Further research to explore the nature of the relationships between the different facets of the conceptual framework in different transdisciplinary contexts in HE, and how this interacts with strength of disciplinary knowledge and the spaces used as well as the pedagogical approaches taken, is needed. It may be the case that normalising a culture of transdisciplinary creativity in HE, STEAM and beyond, will require both physical and

behavioural adjustments regarding course design, pedagogies and space use, where faculty and institutions need to work together to reimagine spaces for teaching and learning.

This study strengthens the growing body of evidence supporting the application, research and development of the creative pedagogies framework in multiple sectors (outlined in the introduction) to date within formal schooling, STEAM initiatives, science education and HE. It also offers insight into the nuances of the development of creative transdisciplinary pedagogy for HE lecturers working in this area which will become important as transdisciplinary practice is increasingly part of Higher Education.

Finally, this research provides a strong framework for moving on understanding and practice in HE communities which are developing transdisciplinary, challenge-based pedagogy with respect to STEAM and beyond. This novel framework, with its depth of insight into how design thinking and creative pedagogies can be brought together for transdisciplinary STEAM HE, offers a framework for further study in STEAM HE educational research that is grounded in an interesting synthesis of theoretical strands of design thinking, creative pedagogies, and transdisciplinarity, and will benefit practitioners as a guide for practice. With an emphasis on contributing to pedagogic design and practice, the project has developed an online fully accessible project toolkit (https://sciculture.eu/coursekit/.) for STEAM lecturers and HE curriculum designers to use this approach in conjunction with the insights into the conceptual framework presented here. The conceptual framework and associated course kit has considerable potential to impact the field further in terms of developing pedagogic practice for transdisciplinary learning based around complex problems and is already being used in Universities across Europe. As HEIs increasingly understand the need to realign their public-facing missions with notions like wicked problems, this research offers a framework that can also be adaptively applied at strategic curricular, as well as course and module design levels.

Note

1. Momigami, or 'kneaded paper', is a technique used within a taught session of course 4, in which participants crumple and uncrumple paper as part of a creative process.

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